page 1 of 6

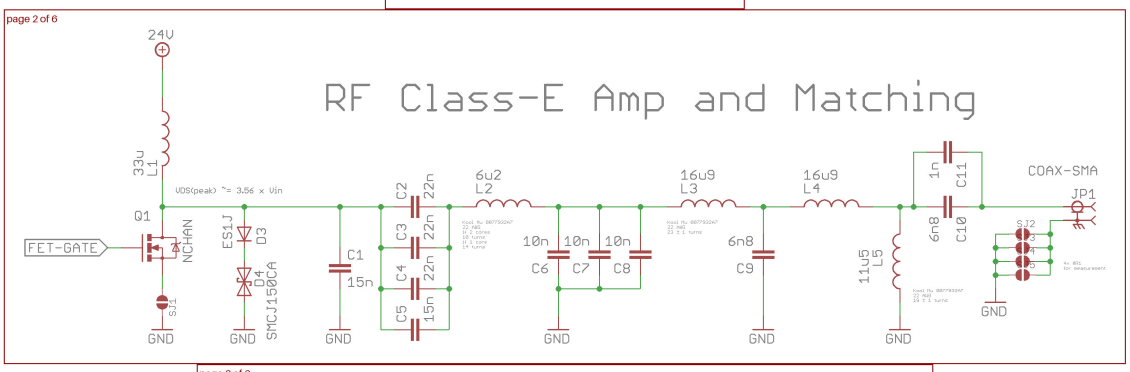
HOT WAND LITE

20260906A

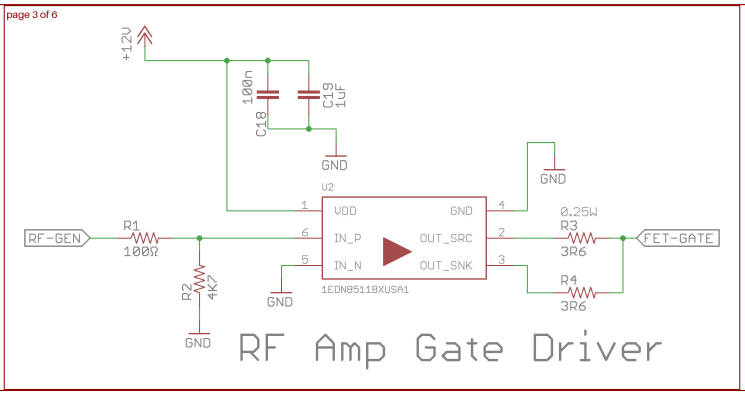
PCB

GND

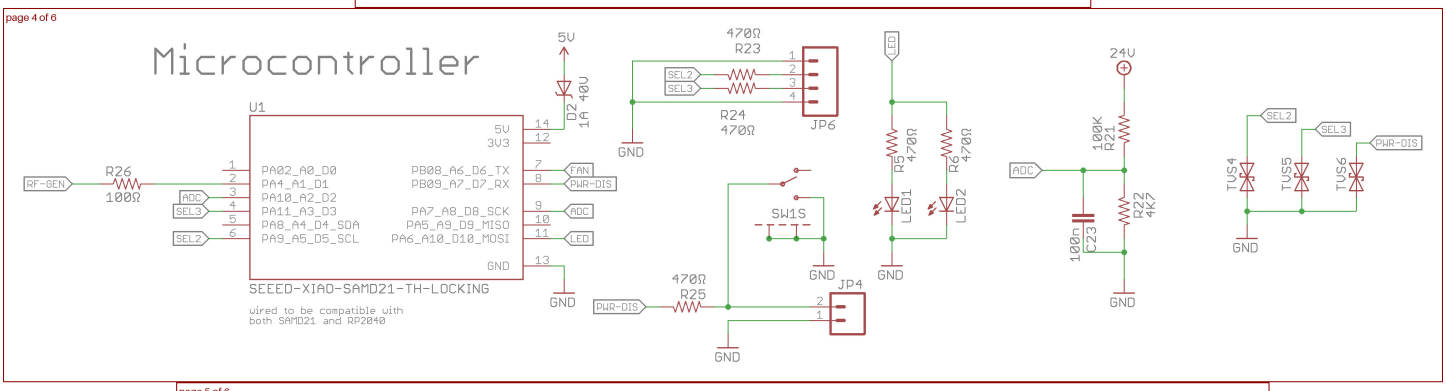
page 2 of 6



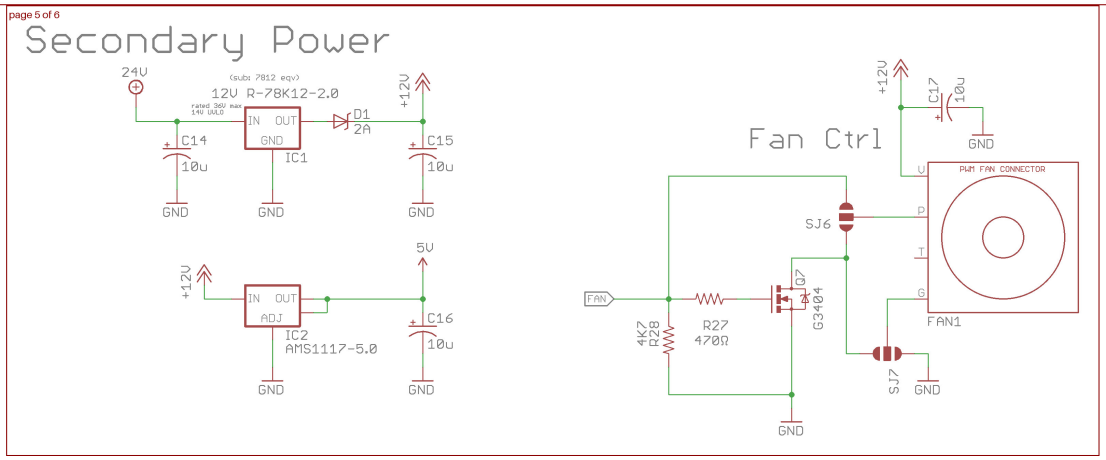
page 3 of 6

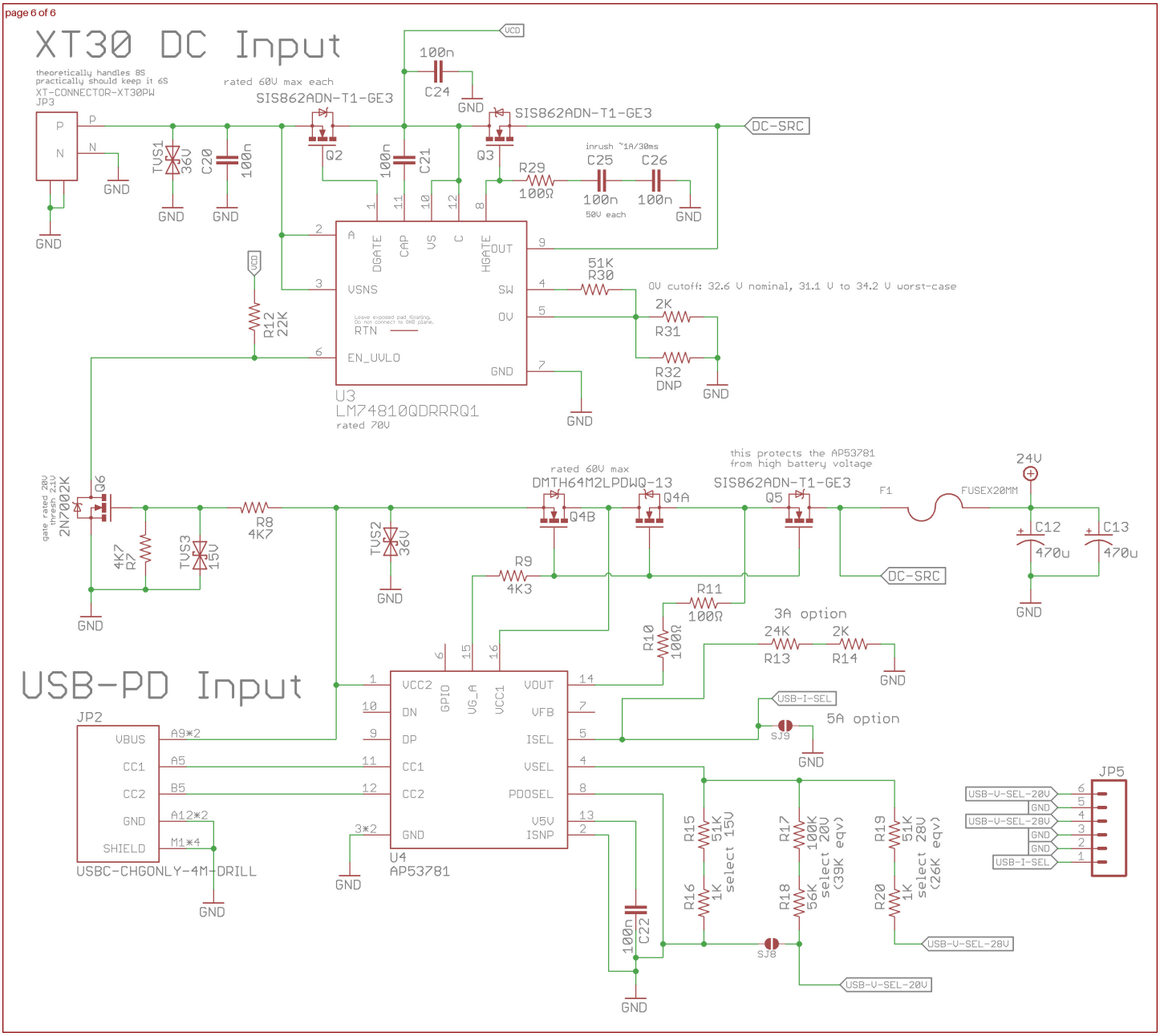


page 4 of 6



page 5 of 6




XT30 DC Input
theoretically handles 8S
practically should keep it 6S
XT-CONNECTOR-XT30PW
JP3
rated 60V max each
SIS862ADN-T1-GE3
Q2
Q3
VCC
100n
C24
GND
DC-SRC
TVS1
36V
C20
100n
C21
R29
100Ω
inrush ~1A/30ms
C25
C26
100n
50V each
U3
LM74810QDRRRQ1
rated 70V
VSNS
RTN
EN_UVLO
DGATE
CAP
VS
C
HGATE
OUT
SW
OV
R12
22K
51K
R30
2K
R31
R32
DNP
OV cutoff: 32.6 V nominal, 31.1 V to 34.2 V worst-case
this protects the AP53781
from high battery voltage
rated 60V max
DMTH64M2LPDWQ-13
Q4B
Q4A
SIS862ADN-T1-GE3
Q5
F1
FUSEX20MM
24V
C12
C13
470u
Q6
2N7002K
gate rated 20V
thresh 2.1V
R8
4K7
R7
4K7
TVS3
15V
TVS2
36V
R9
4K3
R11
100Ω
R10
100Ω
3A option
24K
R13
2K
R14
USB-PD Input
JP2
VBUS
CC1
CC2
GND
SHIELD
A9*2
A5
B5
A12*2
M1*4
USBC-CHGONLY-4M-DRILL
U4
AP53781
VCC2
DN
DP
CC1
CC2
GND
GPIO
VG_A
VCC1
VOUT
VFB
ISEL
VSEL
PDOSEL
V5V
ISNP
3*2
USB-I-SEL
SJ9
5A option
R15
51K
select 15V
R16
1K
R17
100K
select 20V
(39K eqv)
R18
56K
R19
51K
select 28V
(26K eqv)
R20
1K
SJ8
100n
C22
USB-V-SEL-20V
USB-V-SEL-28V
JP5